

Embedding Model Selection and Fine-Tuning

Dense Representation Dimensions

Modern embedding models produce vectors ranging from 384 dimensions (MiniLM) to 1536 (OpenAI text-embedding-3-small) or 3072 dimensions (text-embedding-3-large). Higher dimensions capture finer semantic nuances at higher storage cost.

Domain-Specific Fine-Tuning

Multiple Negatives Ranking Loss (MNRL) and Matryoshka Representation Learning allow fine-tuning embeddings on domain-specific pair datasets (query, positive_doc, negative_doc).

Matryoshka Embeddings

Matryoshka embeddings enable dynamic truncation of vector dimensions (e.g. 1536 -> 256) with under 2% degradation in retrieval accuracy.